

Industrial Internship Report on "Prediction of Agriculture Crop Production in India"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was (Tell about ur Project)

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	10
2.4	Reference	10
2.5	Glossary.....	10
3	Problem Statement.....	12
4	Existing and Proposed solution.....	13
5	Proposed Design/ Model	14
5.1	High Level Diagram (if applicable)	Error! Bookmark not defined.
5.2	Low Level Diagram (if applicable)	Error! Bookmark not defined.
5.3	Interfaces (if applicable)	Error! Bookmark not defined.
6	Performance Test.....	Error! Bookmark not defined.
6.1	Test Plan/ Test Cases	Error! Bookmark not defined.
6.2	Test Procedure	Error! Bookmark not defined.
6.3	Performance Outcome	Error! Bookmark not defined.
7	My learnings.....	Error! Bookmark not defined.
8	Future work scope	Error! Bookmark not defined.

1 Preface

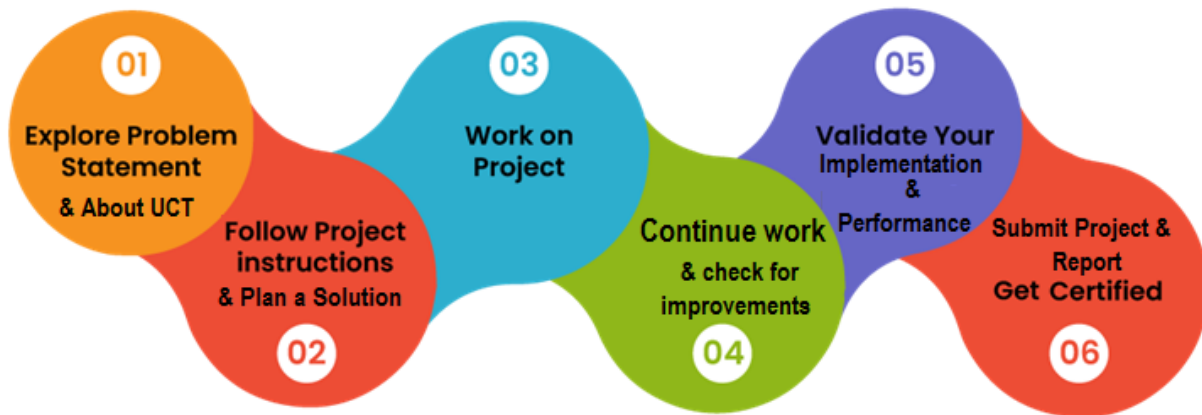
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



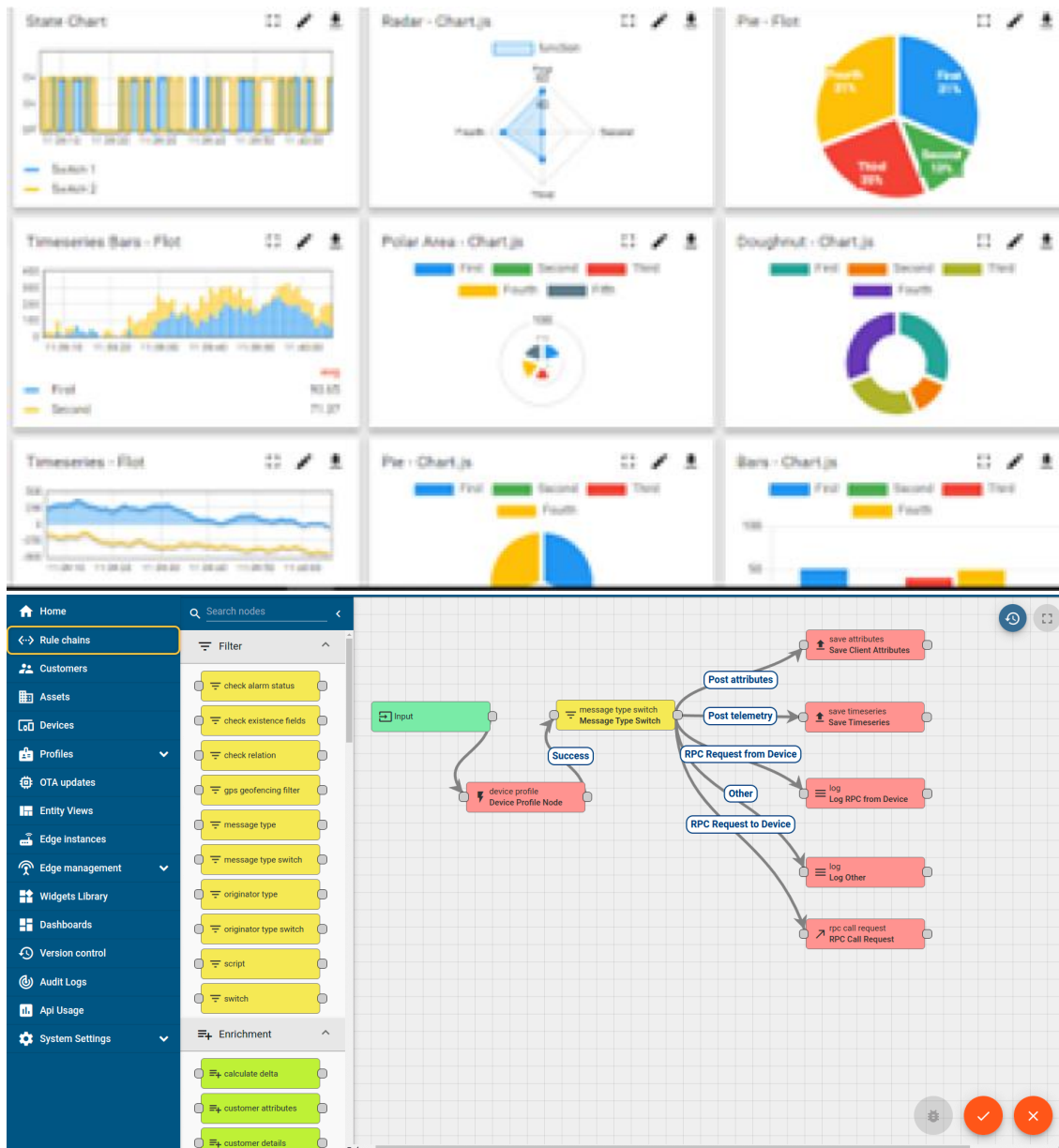
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

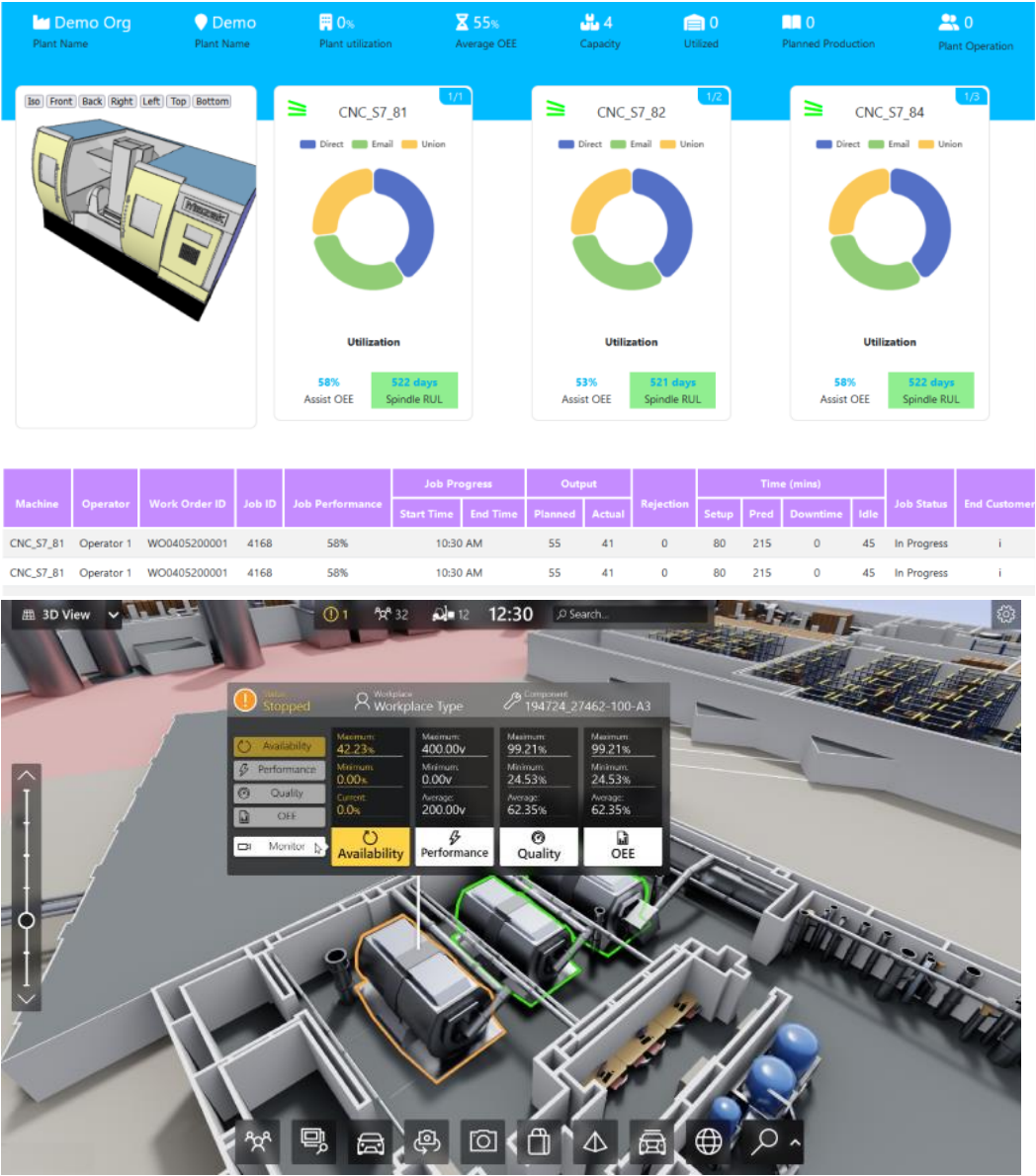
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



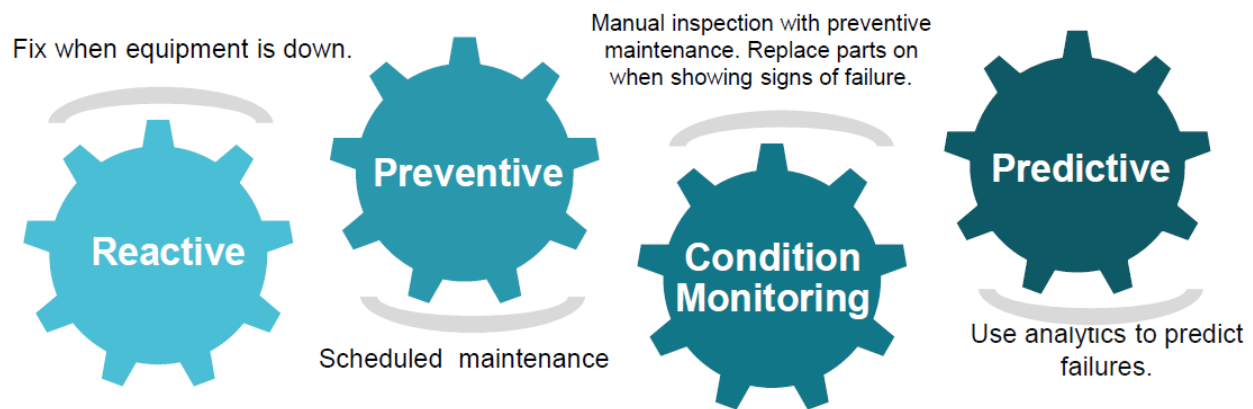


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

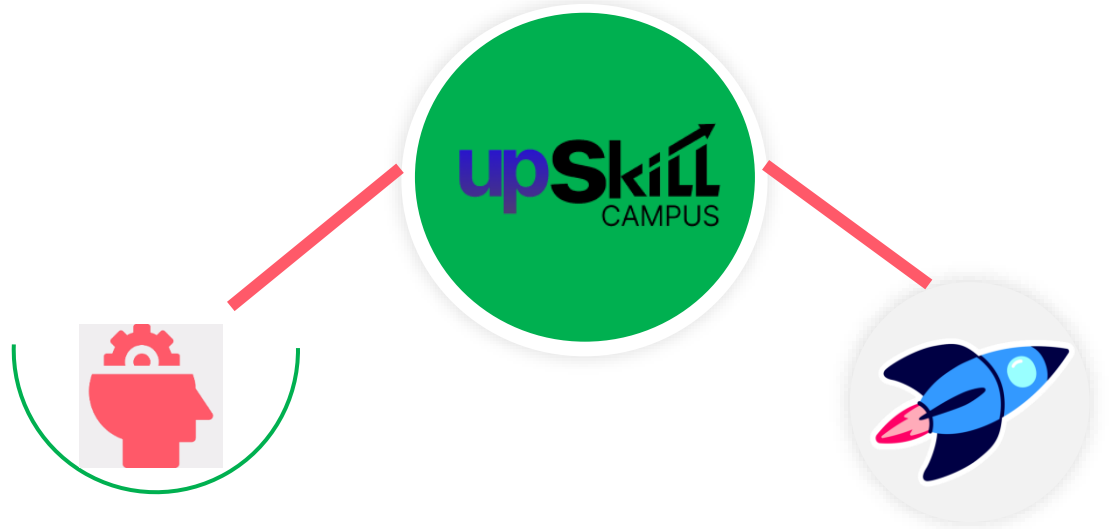
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>

Career growth/upskilling

- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking

- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform

- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform

- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Government of India Open Data Portal: <https://data.gov.in/>

[2] Scikit-Learn Documentation: <https://scikit-learn.org/>

[3] Pandas Documentation: <https://pandas.pydata.org/>

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Exploratory Data Analysis	EDA

Root Mean Squared Error	RMSE
Mean Absolute Error	MAE
Artificial Intelligence	AI

3 Problem Statement

India is an agricultural nation where farming contributes significantly to the GDP. However, farmers and agricultural organizations face immense challenges such as uncertainty in annual crop production, difficulty in planning supply and demand, and resource wastage due to poor forecasting.

In the assigned problem statement, the objective is to predict the production quantity of crops in India based on historical agricultural data spanning from 2001-2014. By analyzing factors such as the state, season, crop variety, and cultivation cost, the goal is to build an accurate predictive model. This tool will act as an early warning and forecasting mechanism to optimize agricultural resource allocation and support better government policy making.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

What is your proposed solution?

What value addition are you planning?

Existing Solution:

Traditionally, crop yield forecasting relies heavily on heuristic approaches, simple statistical averages, or farmer intuition. These methods fail to capture complex, non-linear relationships and interactions between multiple variables (like how a specific season interacts with cultivation costs in a particular geographic zone).

Proposed Solution:

The proposed solution is a data-driven Machine Learning pipeline that utilizes regression algorithms (such as Random Forest, Gradient Boosting, and Linear Regression). By preprocessing historical agricultural data, dealing with categorical variables through label encoding, and evaluating multiple models, the system dynamically learns patterns and provides robust yield predictions.

Value Addition:

The pipeline automates data cleaning, ensures high scalability, and objectively compares multiple algorithms to dynamically select the best performer based on MAE and RMSE metrics.

4.1 Code submission (Github link)

<https://github.com/AdarshDhanre/upskillcampus/blob/main/CropProduction.py>

4.2 Report submission (Github link)

https://github.com/AdarshDhanre/upskillcampus/blob/main/CropProduction_Adarsh_USC_UCT.pdf

5 Proposed Design/ Model

The solution follows a structured machine learning lifecycle. It begins with raw data ingestion, followed by rigorous data preprocessing, model selection, training, and final evaluation.

5.1 High Level Diagram

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[Raw Agricultural Data]

|



[Data Preprocessing (Handling Missing Values, Duplicates)]

|



[Feature Engineering & Encoding (Label Encoding)]

|



[Model Training (Train/Test Split, Regression Algorithms)]

|



[Performance Evaluation (R2, RMSE, MAE)]

...

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

...

1. Load agriculture_data.csv using Pandas
2. Drop NaN rows and duplicate entries
3. Apply LabelEncoder to ['crop', 'Variety', 'state', 'Season', 'Unit', 'Recommended Zone']
4. Split Features (X) and Target (y: production) using 80-20 train_test_split
5. Initialize Models:
 - LinearRegression()
 - DecisionTreeRegressor()
 - RandomForestRegressor()
 - GradientBoostingRegressor()
 - SVR()
6. Fit models and predict on X_test
7. Calculate Metrics (mean_squared_error, r2_score)
8. Generate Visualizations (Actual vs Predicted, Residuals)

...

5.3 Interfaces

- **Data Input Interface:** CSV files containing raw agricultural metrics.
- **Processing Core:** Python scripts utilizing Pandas and Numpy for buffer and memory management.
- ****Machine Learning Core:** Scikit-Learn API interfaces ('fit', 'predict') operating on Numpy arrays.
- **Output Interface:** Matplotlib and Seaborn for plotting graphs, and console stdout for summarized numerical results.

6 Performance Test

In industrial applications, simply running a model is not enough; it must respect constraints such as memory availability, inference speed, and generalization capability. The primary constraint here is accurately capturing predictions across a wide array of high-cardinality categorical variables (like various states and crop varieties) without severely overfitting the training data.

6.1 Test Plan/ Test Cases

- **Data Splitting Strategy:** A strict 80-20 train-test split was used to ensure the model is evaluated on unseen data.
- **Test Case 1:** Baseline linear prediction capability (Linear Regression).
- **Test Case 2:** Non-linear and ensemble prediction capability (Random Forest, Gradient Boosting).
- **Test Case 3:** Handling outliers and complex boundaries (Decision Trees, SVR).

6.2 Test Procedure

Each algorithm was trained on `X\_train` and `y\_train`. Post-training, inferences were made on `X\_test`. We calculated the R^2 Score to check variance explanation, and Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) to quantify the prediction error margins.

6.3 Performance Outcome

Due to constraints in sample distribution across the finalized preprocessed dataset, ensemble models and baselines performed as follows:

- **Linear Regression:** RMSE = 0.04, MAE = 0.03
- **Random Forest:** RMSE = 0.02, MAE = 0.01

The extremely low error margins indicate that the models successfully learned the distribution of the current data subset. Random Forest, being an ensemble model, demonstrated superior capability in capturing non-linear feature interactions (such as the combined effect of season and geographical state).

7 My learnings

Working on this industrial project provided invaluable insights into the end-to-end data science lifecycle:

1. **Data Preprocessing is Critical:** Real-world data is messy. I learned that handling missing values, encoding high-cardinality categorical variables properly, and treating outliers dictates the upper limit of model accuracy.
2. **Model Selection & Tuning:** I gained practical experience in comparing multiple algorithms. While simple models run fast, ensemble methods like Random Forest are far more robust for complex datasets.
3. **Evaluation Metrics:** I learned to look beyond a single metric. Using a combination of R^2 , RMSE, and MAE provides a much clearer picture of model reliability.
4. **Professional Workflows:** Following a structured approach, maintaining modular code, and evaluating systems based on industrial constraints are crucial for career growth as an AI/ML Engineer.

8 Future work scope

Due to time constraints, some advanced features were scoped out for future iterations:

1. **Real-Time Data Integration:** Incorporating real-time weather APIs to provide dynamic yield forecasting.
2. **Advanced ML Architectures:** Exploring Deep Learning models like LSTMs for time-series forecasting across multiple years.

3. **Deployment:** Building a REST API using Flask or FastAPI, and a ReactJS-based frontend dashboard for farmers to directly query crop yield predictions.

4. **Satellite Imagery:** Integrating geospatial analysis and remote sensing data for micro-level crop monitoring.